Invariant-current Taylor chart, principal one-form, and predictable graph recovery

TECT verification-first repository
claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-07-24 · this version issued 2026-07-24 · v1.0

1 Purpose and scope

R-074 isolated a nondecaying bare coefficient resonance and supplied the Besov payment which would close a horizontal cubic payload once a complete adapted one-form with a sixth moment had been constructed. This note makes three rigorous advances and isolates the remaining obstruction more sharply.

First, the production current admits a global, projector-free invariant representation. It gives an exact Taylor chart through the generic, singlet-zero, and collapsed strata and keeps all $A\,DA$ and A^2DU channels. It is an algebraic current representation, not a nondegenerate coordinate chart at the cone tip.

Second, the frozen second-order Taylor coefficient is a canonical R-050 geometric one-form. It converges in every finite probability moment in $C^{-1/2-\kappa}$ and therefore has the sixth moment required by R-074. The negative isolated R-074 coefficient is retained, while the complete three-generator raw-energy coefficient on the same canonical fixture is strictly positive after all restored terms are included. The latter is an attribution oracle, not a global positivity theorem.

Third, for every fixed terminal cutoff, bounded smooth past-cylindrical controls are dense in the declared Cameron–Martin/terminal-L^6 graph domain. The positive raw current, its Wick trace, the retained finite-low term, and the cubic payload are continuous in that topology. Thus a cutoff-uniform endpoint inequality assembled from these graph-continuous terms passes from regular strict-past controls to every control in that graph domain at the same cutoff and with the same constant. Uniform L^6 boundedness of the one-form is exactly sufficient for the cubic pairing, provided the one-form is sequentially identified in probability along the graph recovery sequence.

The full adapted gate does not close. The path coefficient has a genuine third-order transport tail. Its absolute Besov payment is $X^{1/2}Y^{1/2}$ and has no random Young slack. It survives on a non-tip horizontal radial line. Moreover, a constant-control fixture shows that a positive terminal square plus the A^2DA one-form omits a load-bearing $A^2G^{\otimes 2}$ coefficient-curvature/Wick channel. Finally, a smooth adapted coefficient can have infinitely many Wiener chaoses, so R-063's finite unshifted forest cannot simply be relabelled as an arbitrary-adapted theorem.

Accordingly this note does not prove the signed coefficient-transport remainder, the complete adapted gauge-quotient one-form, controlled-shell one-use, the $q = 10/9$ Nelson moment, an interacting measure, floor or regulator removal, or infinite volume. All statements use the fixed positive production floor, common real-even cutoffs, exact non-aliased products, the R-066 finite-low convention, and mutually orthogonal whole-shell strict-past controls. We use $0 < \kappa < 1/2$. For the displayed production decimals, $e = 10^{-12}$ and $P = M_X^2 + \mu_X = 4 + 10^{-12}$; the analytic identities retain e and P symbolically.

2 Production currents and the projector-free invariant chart

Let S_1, S_2, S_3 be the realified Pauli generators. Put

$$\rho = z^T z, \qquad d = \rho + e, \qquad s = \log d, \qquad m_r = z^T S_r z. \tag{2.1}$$

For a spatial derivative $y = D_i z$, define

$$J_{r,i} = D_i m_r, \qquad K_{r,i} = D_i m_r - m_r D_i s. \tag{2.2}$$

The production frames satisfy the exact identity

$$\boxed{M_r(z)^T y = (J_{r,i}, K_{r,i}).} \tag{2.3}$$

Indeed $p_r \cdot y = 2S_r z \cdot y = D_i m_r$, while

$$v_r \cdot y = D_i m_r - \frac{m_r}{d} D_i \rho = D_i m_r - m_r D_i \log d. \tag{2.4}$$

Write

$$Q_{\mathrm{II}} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad a = \frac{9}{500P}, \quad b = \frac{3}{400P}, \quad c = \frac{3}{320P}. \tag{2.5}$$

Set

$$c_0 = \frac{ac - b^2}{c} = \frac{3}{250P}, \qquad c_1 = c\left(1 + \frac{b}{c}\right)^2 = \frac{243}{8000P}, \qquad \alpha = \frac{c}{b+c} = \frac{5}{9}, \tag{2.6}$$

and

$$L_{r,i} = J_{r,i} - \alpha m_r D_i s = d^{\alpha} D_i (m_r d^{-\alpha}). \tag{2.7}$$

Completing the fixed 2×2 quadratic form gives globally

$$\boxed{(J_{r,i}, K_{r,i}) Q_{\mathrm{II}} (J_{r,i}, K_{r,i})^T = c_0 |J_{r,i}|^2 + c_1 |L_{r,i}|^2.} \tag{2.8}$$

In particular $2\alpha = 10/9$, exactly the Nelson exponent already derived by R-064 and R-067.

Both $d\rho$ and every dm_r annihilate the doublet and singlet local phase directions. Equations (2.3)–(2.8) therefore require no phase projector, pseudoinverse, or division by the doublet or singlet amplitude. They remain algebraically valid on every R-073 rank stratum.

There is an essential tip warning. Generically the production frame and $D(\rho, m_1, m_2, m_3)$ both have rank four, and on the singlet-zero stratum both have rank three. At a nonzero pure singlet, however, the production frame has rank zero while $D\rho$ has rank one. Thus (ρ, m) is a global algebraic representation of the currents, not a nondegenerate or bi-Lipschitz quotient coordinate. No inverse estimate at that stratum is used below.

3 Exact invariant-current Taylor remainder

Let U be the uncontrolled endpoint, A a regular control, $G_i = D_i U$, and $C_i = D_i A$. Write

$$\mu_r = 2U^T S_r A, \qquad \eta_r = A^T S_r A, \qquad r_1 = 2U \cdot A, \qquad r_2 = |A|^2, \tag{3.1}$$

$$\Delta s = \log \frac{d + r_1 + r_2}{d}, \qquad R_s = \Delta s - \frac{r_1}{d}. \tag{3.2}$$

Let δJ_r and δL_r denote the complete endpoint current increments and let $\delta J_r^{(1)}, \delta L_r^{(1)}$ denote their Frechet first variations at U in direction A. Direct differentiation gives

$$\boxed{\delta J_r - \delta J_r^{(1)} = D\eta_r,} \tag{3.3}$$

$$\boxed{\delta L_r - \delta L_r^{(1)} = D\eta_r - \alpha\{m_r D R_s + \mu_r D\Delta s + \eta_r D \log(d + r_1 + r_2)\}.} \tag{3.4}$$

Consequently the full raw-energy Taylor remainder after its conservative first variation is exactly

$$\boxed{\begin{aligned}\mathcal{R}_{\mathrm{inv}} = \sum_{r,i}\{&c_0\langle J_{r,i}, \delta J_{r,i} - \delta J_{r,i}^{(1)}\rangle + \frac{c_0}{2}|\delta J_{r,i}|^2 \\ &+ c_1\langle L_{r,i}, \delta L_{r,i} - \delta L_{r,i}^{(1)}\rangle + \frac{c_1}{2}|\delta L_{r,i}|^2\}.\end{aligned}} \tag{3.5}$$

This identity is finite-cutoff algebra. It simultaneously retains the range-visible squares, the phase Ward cancellation, the radial channel, and the cone-tip degeneration. Equations (3.3)–(3.4) show why a completed horizontal formula cannot contain only A^2DA: the exact L remainder also contains $A\,DA$ and A^2DU terms through $D\Delta s$ and DR_s.

For comparison with the earlier frame notation, set

$$M_{r,0} = M_r(U), \quad M_{r,1} = M_r(U + A), \quad D_r = DM_r(U)[A], \quad E_r = M_{r,1} - M_{r,0} - D_r, \tag{3.6}$$

$$w_{r,i} = M_{r,0}^T G_i, \quad d_{r,i} = (D_r + E_r)^T G_i, \quad c_{r,i} = M_{r,1}^T C_i. \tag{3.7}$$

If $F_A(U) = B(U + A) - B(U) - DB(U)[A]$ and $\mathcal{S}_0$ is the R-070 conservative first variation, then

$$\boxed{\begin{aligned}\Delta V_\Lambda^{\mathrm{ren}} - \mathcal{S}_0 = L + N + &\sum_{r,i}\{\langle d_{r,i}, Q_{\mathrm{II}} c_{r,i}\rangle + \tfrac{1}{2}|c_{r,i}|^2_{Q_{\mathrm{II}}}\} \\ &+ \tfrac{1}{2}\langle F_A(U), Q_\Lambda\rangle,\end{aligned}} \tag{3.8}$$

where

$$L = \sum_{r,i}\langle w_{r,i}, Q_{\mathrm{II}} D_r^T C_i\rangle, \qquad N = \sum_{r,i}\langle w_{r,i}, Q_{\mathrm{II}} E_r^T C_i\rangle. \tag{3.9}$$

Equations (3.5) and (3.8) are two exact representations of the same raw endpoint geometry before the A7 Wick conversion. The R-063 lower-chaos chart and the R-066 trace transport must still accompany them.

4 The principal unshifted Taylor one-form

Taylor's integral formula gives

$$E_r(U, A) = \int_0^1 (1 - t) D^2 M_r(U + tA)[A, A], dt. \tag{4.1}$$

Thus the complete E–DA branch has the exact path one-form

$$N = \sum_{i,a,b,c} \langle \mathfrak{I}_{i,abc}^{\mathrm{path}}(U, A), A^a A^b D_i A^c\rangle, \tag{4.2}$$

$$\mathfrak{I}_{i,abc}^{\mathrm{path}}(U, A) = \sum_{r,\mu,\nu,d} M_{r,d\mu}(U) D_i U^d Q_{\mu\nu} \int_0^1 (1 - t)\partial_{ab} M_{r,c\nu}(U + tA), dt. \tag{4.3}$$

Freeze only the Hessian coefficient at the uncontrolled field:

$$\boxed{\mathfrak{I}_{i,abc}^{(2)}(U) = \frac{1}{2}\sum_{r,\mu,\nu,d} M_{r,d\mu}(U) D_i U^d Q_{\mu\nu} \partial_{ab} M_{r,c\nu}(U).} \tag{4.4}$$

The factor 1/2 is $\int_0^1 (1 - t)dt$. At fixed positive floor the tensor

$$\Theta_{abc,d}(u) = \frac{1}{2}\sum_{r,\mu,\nu} M_{r,d\mu}(u) Q_{\mu\nu} \partial_{ab} M_{r,c\nu}(u) \tag{4.5}$$

is smooth and all its derivatives have polynomial growth. Hence $\mathfrak{J}^{(2)} = \Theta(U)DU$ is a finite smooth matrix one-form of the R-071 class.

We record the Holder–Besov strengthening needed here. Choose $\alpha \in (1/3, 1/2)$ with $\alpha > 1/2 - \kappa$. R-050 supplies

$$U \in \mathcal{C}^{\alpha}, \qquad DU \in \mathcal{C}^{\alpha-1}, \qquad U \circ DU \in \mathcal{C}^{-\varepsilon} \tag{4.6}$$

in every finite probability moment, with regulator-independent coupled cutoff convergence. First-order paralinearisation gives

$$\Theta(U) = T_{D\Theta(U)}U + R_{\Theta}, \qquad R_{\Theta} \in \mathcal{C}^{2\alpha}. \tag{4.7}$$

The only nonclassical resonance is reconstructed from $D\Theta(U)(U \circ DU)$ plus the standard commutator. The low–high term has order $\alpha - 1 > -1/2 - \kappa$, and the remaining terms are better. Therefore, for every finite p,

$$\boxed{\mathfrak{J}^{(2)}_{\Lambda} \longrightarrow \mathfrak{J}^{(2)} \quad \text{in } L^p(\Omega; C^{-1/2-\kappa}), \qquad \sup_{\Lambda} \mathbb{E}\|\mathfrak{J}^{(2)}_{\Lambda}\|^6_{C^{-1/2-\kappa}} < \infty.} \tag{4.8}$$

Common real-even covariance makes $\mathrm{Cov}(U, D_iU) = 0$, so no local first-derivative counterterm is created. Since M and D^2M are odd, Θ is even; its Hermite expansion is even and multiplication by DU produces the complete odd one-form chaoses. In particular the first-chaos low–high root is retained.

R-074 now applies exactly:

$$|\langle \mathfrak{J}^{(2)}, A^{\otimes 2} \otimes DA\rangle| \leq \eta X + \zeta Y + C\eta^{-3}\zeta^{-2}\|\mathfrak{J}^{(2)}\|^6_{C^{-1/2-\kappa}}, \tag{4.9}$$

where $X = \|A\|^2_{H^2}$ and $Y = \|A\|^6_6$. This closes the principal unshifted subterm, not the path one-form (4.3).

5 The isolated resonance and its full reassembly

At $z = e_1$ and $A = G = C = e_2$, (4.4) contains the R-074 coefficient

$$-\frac{27}{200P(1+e)}. \tag{5.1}$$

It is nonzero and has no bare shell-separation gain. The complete raw energy on the same local fixture is a stronger attribution test. Put

$$F(t, \ell) = \frac{1}{2}\sum_r |M_r(e_1 + te_2)^T(e_2 + \ell e_2)|^2_{Q_{\mathrm{II}}}. \tag{5.2}$$

The exact $t^2\ell$ coefficient is

$$\boxed{[t^2\ell]F = \frac{3(113e^2 + 136e + 48)}{2000P(1+e)^2} > 0.} \tag{5.3}$$

The σ_x contribution, including both symmetric positions of the second derivative, is

$$-\frac{27}{100P(1+e)}, \tag{5.4}$$

while the restored σ_z transported-linear contribution is

$$\frac{3(113e^2 + 316e + 228)}{2000P(1+e)^2}. \tag{5.5}$$

Their sum is (5.3); σ_y vanishes. At the pinned parameters (5.3) is $0.0180000000000105\ldots$, whereas (5.1) is $-0.0337499999999578\ldots$.

This does not prove global raw or Wick positivity. It proves that the exact R-074 negative branch is only one attribution inside the full signed endpoint and that a verifier which checks only its sign necessarily misses load-bearing restored terms.

6 The coefficient-transport boundary

Subtracting (4.4) from (4.3) gives exactly

$$N = N^{(2)} + N_{\geq 3}, \tag{6.1}$$

$$\boxed{N_{\geq 3} = \frac{1}{2}\sum_{r,i}\int_0^1 (1-s)^2 \left\langle M_r(U)^T G_i, Q_{\mathrm{II}}[D^3 M_r(U+sA)[A,A,A]]^T C_i \right\rangle ds.} \tag{6.2}$$

At fixed floor D^3M is globally bounded. The natural endpoint product estimate is

$$\|A^{\otimes 3} \otimes DA\|_{B^{1/2+\kappa}_{1,1}} \leq C_\kappa \|A\|_{H^2}\|A\|_6^3. \tag{6.3}$$

The proof is the R-074 largest-frequency argument with the three undifferentiated factors in L^6. Pairing an order $C^{-1/2-\kappa}$ random current produces

$$R(\omega)X^{1/2}Y^{1/2}. \tag{6.4}$$

The deterministic exponents already sum to one. An inequality

$$R\sqrt{XY} \leq \eta X + \zeta Y + C_R \tag{6.5}$$

for all $X, Y \geq 0$ requires $R \leq 2\sqrt{\eta\zeta}$; an additive constant cannot repair the large common scaling of X and Y. An unbounded random current therefore has no absolute Young route, even if it has every finite moment. This scoped failure is registered as `NG-2026-07-24-A13-ABSOLUTE-THIRD-ORDER-TRANSPORT`.

The tail is not a phase artifact. On the active-real σ_3 radial line take

$$U = A = \sqrt{e}\, e_1, \qquad G = C = e_1. \tag{6.6}$$

There $p(u) = 2u$ and $v(u) = 2eu/(u^2+e)$. With $f(r) = 2r/(1+r^2)$,

$$f(1) = 1, \quad f(2) = 4/5, \quad f'(1) = 0, \quad f''(1) = -1. \tag{6.7}$$

Hence

$$E_v = -0.2\sqrt{e}, \qquad \frac{1}{2}D^2 v[A,A] = -0.5\sqrt{e}, \qquad E_{3,v} = 0.3\sqrt{e}, \tag{6.8}$$

and

$$\boxed{N_{\geq 3} = 0.3e(2q_{12} + q_{22}) = \frac{0.0073125\, e}{P} > 0.} \tag{6.9}$$

At the pinned production $P = 4 + 10^{-12}$ this is approximately $0.001828125\, e$. This is a non-tip horizontal direction; neither local phase invariance nor terminal-kernel cancellation deletes it.

Also, $\mathfrak{J}^{(2)}$ is not individually gauge-horizontal. An independent constant phase-tangent fixture gives a nonzero principal contraction while the exact rotated raw energy is invariant. The Ward cancellation uses the orbit's second-order radial acceleration, both R-073 restored first variations, (6.2), and the terminal square. Calling (4.4) the completed gauge-quotient one-form would therefore be false.

7 Two omitted-channel diagnostics

7.1 Constant-control coefficient curvature

Let

$$F(z,y) = \frac{1}{2}\sum_r |M_r(z)^T y|^2_{Q_{\mathrm{II}}}. \tag{7.1}$$

For fixed z, a, y, define

$$R_0 = F(z+a,y) - F(z,y) - D_zF(z,y)[a]. \tag{7.2}$$

With

$$\delta j_r = [M_r(z+a) - M_r(z)]^T y, \qquad E_r = M_r(z+a) - M_r(z) - DM_r(z)[a], \tag{7.3}$$

ordinary algebra gives

$$\boxed{R_0 = \frac{1}{2}\sum_r |\delta j_r|_Q^2 + \sum_r (M_r(z)^T y)^T Q_{\mathrm{II}}(E_r^T y).} \tag{7.4}$$

On the independently found production fixture recorded by the executables,

$$R_0 = -0.01977052360318\ldots, \qquad \frac{1}{2}\sum_r |\delta j_r|_Q^2 = 0.01790632256879\ldots, \tag{7.5}$$

$$\sum_r (M_r(z)^T y)^T Q(E_r^T y) = -0.03767684617197\ldots. \tag{7.6}$$

Here $C = DA = 0$, so every A^2DA one-form vanishes. The negative coefficient-curvature channel is larger than the retained square. Therefore "positive square plus the cubic one-form" is not the full endpoint. The $A^2G^{\otimes 2}$ Wick channel, R-063 lower chaoses, and exact trace transport must remain. This is registered as `NG-2026-07-24-A13-ONEFORM-ONLY-ENDPOINT-OMISSION`.

7.2 An adapted coefficient can have an infinite forest

For $X \sim N(0,1)$ consider the smooth R-074 feedback factor

$$f(X) = e^{-X^2}(X^2 - 1). \tag{7.7}$$

If H_n are the probabilists' Hermite polynomials and $c_n = \mathbb{E}[fH_n]/n!$, the generating function is

$$\sum_{n\geq 0} c_n t^n = \frac{e^{-t^2/3}}{\sqrt{3}}\left(\frac{t^2}{9} - \frac{2}{3}\right). \tag{7.8}$$

Thus

$$c_0 = -\frac{2}{3\sqrt{3}}, \qquad c_{2m} = \frac{(-1)^{m-1}(m+2)}{\sqrt{3}\,3^{m+1}m!} \neq 0 \quad (m \geq 1). \tag{7.9}$$

This smooth terminal adapted factor therefore has infinitely many chaoses, so an arbitrary adapted substitution need not have a finite forest. R-063's exact P_3/P_1 and $P_4/P_2/\Sigma Q/P_0$ forest remains the correct unshifted and deterministic- translation chart, but it is not automatically complete after a correlated adapted substitution. A viable proof must causal-freeze controls relative to each fresh root or prove an arbitrary-multiplier/shifted-enhancement theorem. This failure is registered as `NG-2026-07-24-A13-ADAPTED-FINITE-CHAOS-TRANSFER`.

8 Predictable fixed-cutoff graph recovery

Fix $J < \infty$. For the mutually orthogonal strict sharp-cube controls put

$$a_j = K_j h_j, \qquad h_j \in L^0(\mathcal{F}_{j-1}; \mathcal{H}_j), \qquad S_J h = \left(\sum_{j\leq J} |a_j|^2\right)^{1/2}. \tag{8.1}$$

Define

$$\mathcal{A}^2_{J,6} = \left\{ h : \ \mathbb{E}\sum_{j\leq J} \|h_j\|^2_{\mathcal{H}_j} + \mathbb{E}\|S_J h\|_6^6 < \infty \right\}, \tag{8.2}$$

with graph norm

$$\|h\|_{\mathcal{G}_J} = \left(\mathbb{E}\sum_j \|h_j\|_{\mathcal{H}_j}^2\right)^{1/2} + \left(\mathbb{E}\|S_J h\|_6^6\right)^{1/6}. \tag{8.3}$$

Sharp-shell Littlewood–Paley equivalence identifies the second coordinate with the terminal $L^6(\Omega; L_x^6)$ control norm. The shell control map also gives

$$\mathbb{E}\left\|\sum_j K_j(h_j - g_j)\right\|_{H^2}^2 \lesssim \mathbb{E}\sum_j \|h_j - g_j\|_{\mathcal{H}_j}^2. \tag{8.4}$$

At fixed J, each shell and its canonical Gaussian past are finite dimensional; the declared filtration carries no additional unapproximated auxiliary randomness. Shell projection and inversion show that terminal L^6 control implies each h_j belongs to $L^6(\Omega; \mathcal{H}_j)$. Radially truncate h_j itself, which preserves $\mathcal{F}_{j-1}$-measurability, then approximate the bounded function of the finite Gaussian past by a smooth cylinder. Doing this shell by shell produces bounded smooth past-cylindrical $h^{(n)}$ with

$$\boxed{\|h^{(n)} - h\|_{\mathcal{G}_J} \longrightarrow 0.} \tag{8.5}$$

No terminal-event truncation is used.

It remains to identify the positive current graph. Put

$$\mathbf{F}_e(u) = (Q_{\mathrm{II}}^{1/2} M_r(u)^T \partial_i u)_{r,i}. \tag{8.6}$$

The production frame is globally Lipschitz and has linear growth:

$$|M_r(u)| \le C|u|, \qquad |M_r(u) - M_r(v)| \le C|u - v|. \tag{8.7}$$

Indeed $|q| \le 1$ and $|z||Dq(z)[w]| \le 4|w|$, so $|Dv(z)[w]| \le 12|w|$, uniformly over positive floors. The interpolation

$$\|\nabla u\|_3 \le C\|u\|_6^{1/2}\|u\|_{H^2}^{1/2} \tag{8.8}$$

implies

$$\|\mathbf{F}_e(u)\|_2 \le C\|u\|_6^{3/2}\|u\|_{H^2}^{1/2}. \tag{8.9}$$

For $d = u - v$,

$$\begin{aligned}\|\mathbf{F}_e(u) - \mathbf{F}_e(v)\|_2 \le C\{&\|d\|_6\|u\|_6^{1/2}\|u\|_{H^2}^{1/2} \\ &+ \|v\|_6\|d\|_6^{1/2}\|d\|_{H^2}^{1/2}\}.\end{aligned} \tag{8.10}$$

Holder in probability, with exponents $(3, 6, 2)$ in the first term and $(3, 6, 2)$ in the second, proves

$$\mathbf{F}_e(U_J + A_n) \longrightarrow \mathbf{F}_e(U_J + A) \quad \text{in } L^2(\Omega; L_x^2). \tag{8.11}$$

Hence the positive raw current converges in $L^1(\Omega)$. At fixed cutoff the Wick trace satisfies

$$|T_J(u) - T_J(v)| \le C_J(\|u\|_2 + \|v\|_2)\|u - v\|_2, \tag{8.12}$$

so the covariance-normal endpoint and the retained finite-low endpoint also converge in L^1. Any endpoint lower bound assembled from these graph-continuous production terms and whose constant is uniform in J therefore passes from regular controls to every $h \in \mathcal{A}_{J,6}^2$ with the same constant. The approximation constants may depend on fixed J; no removal limit is asserted.

Plain predictable L^2 density is insufficient. If $E_n \in \mathcal{F}_{j-1}$ has probability $1/n$ and

$$h_n = n^{1/6}\mathbf{1}_{E_n} K_j^{-1} f, \tag{8.13}$$

then

$$\mathbb{E}\|h_n\|^2_{\mathcal{H}_j} = n^{-2/3}\|K_j^{-1}f\|^2_{\mathcal{H}_j} \to 0, \qquad \mathbb{E}\|K_j h_n\|_6^6 = \|f\|_6^6. \tag{8.14}$$

This registers `NG-2026-07-24-A13-L2-ONLY-PREDICTABLE-RECOVERY` and explains why the sextic graph coordinate in (8.3) is necessary.

Finally, polarization of R-074 gives

$$\begin{aligned}\|A^2DA - B^2DB\|_{B^{1/2+\kappa}_{1,1}} \lesssim{}& \|A-B\|_{H^2}(\|A\|_6+\|B\|_6)^2 \\ &+ \|A-B\|_6(\|A\|_6+\|B\|_6)(\|A\|_{H^2}+\|B\|_{H^2}).\end{aligned} \tag{8.15}$$

Thus the payload converges strongly in $L^{6/5}(\Omega; B^{1/2+\kappa}_{1,1})$. Uniform L^6 boundedness of an adapted one-form, together with convergence in probability along (8.5), is exactly enough to pass its expected pairing: truncate the one-form norm, use probability convergence on the bounded part, and use L^6–$L^{6/5}$ duality on the payload tail. No $6+\varepsilon$ moment is needed. Moment boundedness alone is not graph identification: an alternating sequence $(-1)^n\mathfrak{J}_0$ has all moments but no pairing limit.

9 Evidence map and successor

The proof map after R-075 is:

1. **Closed algebraically:** (2.3)–(3.5) give the complete projector-free invariant-current Taylor chart, with the tip degeneration stated rather than inverted.
2. **Closed analytically:** the principal unshifted tensor (4.4) is a canonical $C^{-1/2-\kappa}$ R-050 one-form with every finite moment and the required cutoff-uniform sixth moment. R-074 pays its A^2DA pairing.
3. **Closed as attribution:** (5.3)–(5.5) reassemble the negative isolated resonance into the positive complete coefficient on the canonical fixture. This is not a global sign theorem.
4. **Closed negatively:** the absolute $N_{\geq 3}$ route is Young-critical; a horizontal radial fixture proves the tail survives.
5. **Closed negatively:** a one-form-only endpoint omits the coefficient-curvature/Wick channel, and a smooth adapted counterexample shows that arbitrary adapted substitution need not preserve a finite R-063 forest.
6. **Closed conditionally on the regular endpoint inequality:** the predictable fixed-cutoff graph recovery extends that inequality to the declared graph domain, and uniform L^6 boundedness plus graph identification passes the cubic pairing.
7. **Still open:** reconstruct and bound $N_{\geq 3}$ together with the R-063 Wick/lower-chaos channel, both R-073 restored first variations, the terminal square, and the finite-low boundary by one signed causal identity.

The smallest honest successor is

$$\boxed{\texttt{A13-CLASSII-SIGNED-COEFFICIENT-TRANSPORT-ENDPOINT-REMAINDER.}} \tag{9.1}$$

It may close through a signed Ward/martingale telescope or an adapted arbitrary-multiplier shifted enhancement. It may not use a termwise absolute third-order Besov payment, call the principal tensor gauge-complete, freeze a terminal adapted coefficient as a finite chaos, omit the $C=0$ Wick channel, or use a singular phase projector. Once it closes for regular controls, Section 8 removes the separate fixed-cutoff density obstruction; the umbrella one-use and $q=10/9$ Nelson synthesis then remain.

10 Devil's-advocate review

1. **"The invariant variables are a smooth global quotient coordinate." UPHELD as false.** The pure-singlet rank is zero for the production frame and one for $D\rho$. Only the projector-free current identity is asserted.

2. **"The principal tensor is the completed horizontal one-form." UPHELD as false.** It is nonzero on a constant phase tangent; Ward completion needs the second-order orbit acceleration, restored first variations, $N_{\geq 3}$, and the terminal square.

3. **"The R-074 negative coefficient survives with the same sign in the full canonical fixture." DISMISSED.** Exact reassembly gives the positive coefficient (5.3). No global positivity is inferred.

4. **"Every-moment control of the transported current repairs the third-order estimate." UPHELD as false for the absolute route.** The deterministic powers in (6.4) already exhaust Young's inequality.

5. **"The terminal square makes the $C = 0$ channel harmless." UPHELD as false.** The fixture (7.5)–(7.6) has a negative curvature pair larger than the positive square.

6. **"R-063's finite forest remains finite after arbitrary adapted substitution." UPHELD as false.** Equations (7.8)–(7.9) give infinitely many nonzero even chaoses.

7. **"Ordinary predictable L^2 density is enough." UPHELD as false.** The exact spike (8.13)–(8.14) loses the terminal sextic coordinate. The graph norm (8.3) repairs it.

8. **"Graph recovery needs a $6 + \varepsilon$ one-form moment." DISMISSED.** Strong $L^{6/5}$ payload convergence and uniform L^6 boundedness suffice by truncation, provided the one-form is identified in probability.

9. **"This closes one-use or Nelson." UPHELD as an overclaim.** The signed coefficient-transport endpoint remainder, umbrella one-use, and Nelson synthesis remain open.

11 Reproduction and independent audit

The primary executable derives all production constants from A1, verifies the invariant-current and Nelson-aligned diagonalizations, the exact Taylor chart, two-resolution principal/path split, full resonance reassembly, constant-control omission, horizontal radial tail, exact Hermite generating function, frame Lipschitz bounds, critical exponent ledger, predictable L^2 spike, and a spectral graph-recovery sequence. The non-importing executable rebuilds the complex Pauli frames, uses fourth-order stencils, two-dimensional resonance extraction, two-resolution Gauss–Hermite quadrature, a different graph fixture, and the same exact omission oracle.

```
python codes/foundations/
  a13_classii_principal_taylor_oneform_graph_recovery_verify.py
```

Result footer

```
Result ID: A13-CLASSII-PRINCIPAL-TAYLOR-ONE-FORM-GRAPH-RECOVERY-REDUCTION
Precise statement:       The production current has an exact projector-free
                         invariant Taylor chart. Its principal unshifted
                         second-order one-form is canonical in every finite
                         Lp(C^(-1/2-kappa)) with a cutoff-uniform sixth
```

	moment. Fixed-cutoff predictable graph recovery is valid. The signed coefficient-transport tail remains.
Scope:	Fixed-floor A1/R-050/R-063/R-071/R-073/R-074; common real-even cutoffs; regular whole-shell then finite-energy strict-past controls at each fixed J; finite-low retained.
Dependencies:	A1; R-050; R-063; R-066--R-074.
Evidence grade:	ANALYTIC, EXACT, EXECUTED.
Reproduction command:	python codes/foundations/ a13_classii_principal_taylor_oneform_graph_recovery_verify.py
Expected output:	primary + independent + integrated aggregate PASS.
Falsification gate:	An invariant-current, Taylor, resonance, omission, radial-tail, Hermite, graph, hash, PDF, or scope assertion fails.
Tier before / after:	T4 / T4; no promotion.
No-overclaim statement:	No completed adapted gauge-quotient one-form, signed coefficient transport, one-use, Nelson, measure, removal, infinite volume, or T5--T7.
Next required action:	Prove the signed coefficient-transport endpoint remainder with the R-063 lower chaoses, restored first variations, terminal square, and finite-low boundary retained in one causal identity.